UNLOX Minor Project - 5

PROJECT REPORT: TELCO CUSTOMER CHURN PREDICTION USING BOOSTING

1. OVERVIEW & DATASET

NAME: Vansh Modi
DATASET: Telco Customer Churn
DATASET LINK: https://www.kaggle.com/datasets/blastchar/telco-customer-churn
SIZE: 7,043 rows & 21 columns
GOAL: Predict customer churn (Yes/No) using Boosting algorithms (AdaBoost, Gradient Boosting, XGBoost) and identify key business drivers of customer attrition.

2. DATA CLEANING & PREPROCESSING SUMMARY

- Removed non-informative **customerID**.
- Identified a data-type discrepancy in **TotalCharges** (stored as text due to whitespace **" "** in 11 rows where **tenure** == **0**).
- Converted the column to **float64** & imputed the 11 zero-tenure rows with **0.0**.
- Engineered **Avg_Monthly_Difference** feature, applied One-Hot Encoding, and resolved the 73.5% vs. 26.5% class imbalance using **SMOTEENN** (Synthetic Minority Over-sampling + Edited Nearest Neighbours).

3. KEY EXPLORATORY INSIGHTS

1. **Imbalanced Target:** 26.5% of customers churned while 73.5% stayed in the raw data.
2. **Contract Sensitivity:** Month-to-month contract holders account for the vast majority of churn.
3. **Early Churn Risk:** Churn is highest within the first 12 months of tenure and among customers with high monthly bills ($70–$110).
4. **Service Friction:** Fiber optic internet users churn at much higher rates than DSL users.

4. METHODOLOGY & MODEL RESULTS

I evaluated three sequential ensemble Boosting models on the SMOTEENN-processed dataset:

- **AdaBoost:** ~90.4% Accuracy | ~0.965 ROC-AUC
- **Gradient Boosting:** ~93.5% Accuracy | ~0.982 ROC-AUC
- **XGBoost: ~94.2% Accuracy | ~0.985 ROC-AUC | ~95% Recall**

5. CONCLUSION

Boosting algorithms combined with SMOTEENN class balancing effectively capture non-linear interactions between contract type, tenure, and billing. Deploying the XGBoost model allows the retention team to proactively target over 95% of high-risk customers during their first year.